

LAB RESOURCE UTILIZATION PLATFORM

Document 4

Functional and Non-Functional Requirements

Infosys Springboard Internship

1. Introduction

The functional requirements define the services provided by the Lab Resource Utilization Platform, while the non-functional requirements specify the quality attributes that the system must satisfy.

2. Functional Requirements

Requirement ID	Requirement	Description
FR-01	User Authentication	The system shall provide secure user login and logout.
FR-02	User Management	The system shall allow administrators to manage users and roles.
FR-03	Equipment Management	The system shall allow adding, updating, deleting, and viewing equipment.
FR-04	Equipment Booking	Users shall be able to reserve available equipment.
FR-05	Booking Approval	Lab Managers shall approve or reject booking requests.
FR-06	Maintenance Management	The system shall maintain equipment maintenance records.
FR-07	Resource Sharing	The system shall support resource sharing between institutions.
FR-08	Notifications	The system shall notify users about bookings and maintenance.
FR-09	Reports	The system shall generate utilization and maintenance reports.

3. Non-Functional Requirements

Requirement ID	Requirement	Description
NFR-01	Performance	The system shall respond to user requests within acceptable time.
NFR-02	Security	The system shall use secure authentication and role-based access.
NFR-03	Availability	The platform shall be available whenever required.
NFR-04	Reliability	The system shall provide accurate and consistent data.
NFR-05	Scalability	The system shall support future growth.
NFR-06	Usability	The interface shall be simple and user-friendly.
NFR-07	Maintainability	The system shall support easy maintenance and updates.
NFR-08	Compatibility	The application shall work on modern web browsers.

4. Conclusion

The above functional and non-functional requirements define the expected behavior, quality, and performance of the Lab Resource Utilization Platform and provide the foundation for system design and implementation.